

Chapter 3

Management of Medication (MOM)

Intent of the chapter

The organisation has a safe and organised medication management process. The availability, safe storage, prescription, dispensing and administration of medications is governed by written guidance. The organisation designates a medication safety officer.

The organisation develops, implements and updates the hospital formulary. The pharmacy shall have oversight of all medications stocked out of the pharmacy. The pharmacy shall ensure correct storage (as regards to temperature, light; high-risk medications including look-alike, sound-alike, etc.), expiry dates and maintenance of documentation.

The availability of emergency medication is stressed upon. The organisation shall have a mechanism to ensure that the emergency medications are standardised throughout the organisation, readily available and replenished promptly. There shall be a monitoring mechanism to ensure that the required medications are always stocked and well within expiry dates.

Every high-risk medication order shall be verified by an appropriate person to ensure accuracy of the dose, frequency and route of administration. Safety is paramount when using narcotics, chemotherapeutic agents and radiopharmaceuticals. Reconciliation of medications occurs at transition points of patient care as part of patient safety.

The medication management process also includes monitoring of patients after administration and procedures for reporting and analysing near-misses, medication errors and adverse drug reactions.

Medications also include blood, implants and devices.

Medical supplies and consumables are available for use.

SUMMARY OF STANDARDS

MOM.1.	Pharmacy services and medication management is done safely.
MOM.2.	The organisation develops, updates and implements a hospital formulary.
MOM.3.	Medications are stored appropriately and are available where required.
MOM.4.	Medications are prescribed safely and rationally.
MOM.5.	Medication orders are written in a uniform manner.
MOM.6.	Medications are dispensed in a safe manner.
MOM.7.	Medications are administered safely.
MOM.8.	Patients are monitored after medication administration.
MOM.9.	Narcotic drugs and psychotropic substances, chemotherapeutic agents and radio-pharmaceuticals are used safely.
MOM.10.	Implantable prosthesis and medical devices are used in accordance with laid down criteria.
MOM.11.	Medical supplies and consumables are stored appropriately and are available where required.

*This implies that the objective element requires documentation.

Summary of Objective Elements

Standard	11
Objective elements	68
CORE	13
Commitment	48
Achievement	6
Excellence	1

Objective Element	MOM.1.	MOM.2.	MOM.3.	MOM.4.	MOM.5.	MOM.6.	MOM.7.	MOM.8.	MOM.9.	MOM.10.	MOM.11.
a	Commitment	CORE	CORE	Commitment	Commitment	Commitment	Commitment	Commitment	Commitment	Commitment	Commitment
b	Commitment	Commitment	Commitment	CORE	Commitment	Commitment	Commitment	Commitment	Commitment	Commitment	Commitment
c	Achievement	Commitment	CORE	Commitment	Commitment	Commitment	Commitment	CORE	Commitment	Commitment	Commitment
d	Commitment	Achievement	Achievement	Excellence	Commitment	CORE	CORE	Commitment	Commitment	Commitment	Commitment
e	Commitment	Commitment	CORE	CORE		CORE	Commitment	Commitment	Commitment	Achievement	Commitment
f		Commitment	Commitment	CORE		Commitment	Commitment	Commitment			
g			CORE	Achievement			Commitment				
h				Achievement			CORE				
i							Commitment				
j							Commitment				
k							Commitment				

Standards and Objective Elements

Standard

MOM.1. Pharmacy services and medication management is done safely.

Objective Elements

Commitment a. Pharmacy services and medication management are implemented following written guidance. *

Interpretation: The written guidance shall address aspects related to the formulary, procurement, storage, prescription, dispensing, administration and monitoring the use of medications. A qualified individual shall supervise all the activities of the Pharmacy services. Medication management shall cover all patient care area across the organisation. All required procedures under this chapter shall be documented as a 'Medication Management Manual'.

Commitment b. A multi-disciplinary committee guides the formulation and implementation of pharmacy services and medication management.

Interpretation: The multidisciplinary committee shall have defined roles and responsibilities for the management of medications. Where applicable, these are in line with legislation and applicable regulations. Some of the responsibilities of the committee include developing medication management processes; developing and revising the organisation's formulary; and evaluating medication use, and patient safety incidents involving medications. The committee shall include representatives of major clinical departments, administration, a pharmacist/clinical pharmacologist and medication safety officer. It shall also have representation from nurses and Quality department. The objectives of this committee, terms of reference like its composition, frequency of meetings, the quorum required shall be defined. At a minimum, the committee shall meet once in three months. The minutes of the meeting shall be documented.

Achievement c. The multi-disciplinary committee updates medication management processes.

Interpretation: This includes information on rational use, medication errors, medication management, adverse drug reactions, patient safety and high-risk

medications. The multidisciplinary committee shall conduct Medication Management system review on an annual basis. The organization designates a medication safety officer, this job responsibility could be handled by the patient safety officer for example monitoring and analysis of medication related adverse events.

Refer PSQ.1.d. and PSQ.7.c.

Commitment d. There is a procedure to obtain medications when the pharmacy is closed or in case of stock outs. *

Interpretation: When the pharmacy is closed or in case of stock outs there shall be a standard operating procedure to procure the drugs. It is preferable that the organisation has a 24-hour pharmacy.

Commitment e. The organisation has a mechanism to inform relevant staff of key changes in pharmacy services and medication management to ensure uninterrupted and safe care.

Interpretation: The organisation shall have a process to communicate medication shortages, including stockouts to relevant staff (clinicians and nurses). The relevant staff shall also be informed of the recall of a drug within 24 hours, any serious adverse events and patient safety incidents associated with medication usage. The Pharmacy-in-charge or a designated authority shall be responsible for communicating the same. This Communication shall be documented.

Refer to MOM.2., MOM.5.b. and MOM.9.e.

Standard

MOM.2.

The organisation develops, updates and implements a hospital formulary.

Objective Elements

CORE

a. A list of medications appropriate for the patients and as per the scope of the organisation's clinical services is developed collaboratively by the multi-disciplinary committee.



CORE



Commitment



Achievement



Excellence

Interpretation: The multidisciplinary committee shall prepare the organisation's formulary. The formulary shall include medications necessary to meet the organisation's mission, patient needs and scope of services. The formulary could be prepared keeping in view the "National List of Essential Medicines" and "WHO Model List of Essential Medicines". It could also take into consideration the harm potential of the medication, interactions with other medications and likelihood of patient safety incidents. The organisation could look at the possibility of having system-wise/speciality wise formulary.

At a minimum, the formulary shall include the name of the molecule, formulation and strength(s). The organisation shall endeavour to limit the number of drug concentrations of a particular drug in the formulary.

Implants and devices are included in drugs and shall be included in the formulary.

Commitment b. The list is reviewed and updated collaboratively by the multi-disciplinary committee at least annually.

Interpretation: The committee may review and update all medications or only certain medication categories. The review may be done speciality wise. Non-formulary drugs, which were procured in the previous year regularly, may be included in the revised list. Aspects of patient safety, including adverse drug reactions, changing disease pattern, changing resistance pattern, and cost, could be taken into consideration during the review.

Commitment c. The current formulary is available for clinicians to refer to.

Interpretation: The current formulary shall be made available to all treating doctors of the organisation. The organisation needs to ensure that clinicians have access to the current version of the formulary. The formulary could be made available in either physical or electronic form.

Achievement d. The clinicians adhere to the current formulary.

Interpretation: The organisation shall ensure that prescriptions are as per the formulary. It shall monitor the frequency of prescriptions being rejected/local purchase done because it contained non-formulary drugs.

- Commitment e. The organisation adheres to the procedure for the acquisition of formulary medications. ***

Interpretation: The procedure shall address the issues of vendor selection, vendor evaluation, reorder levels, indenting process, generation of the purchase order, and receipt of goods. The procedure also addresses managing stock-outs due to various reasons.

- Commitment f. The organisation adheres to the procedure to obtain medications not listed in the formulary. ***

Interpretation: Local purchase/hotline, which takes care of the immediate requirement are examples of the procedure to obtain medications not listed in the formulary. Whenever there is the local purchase of a medication that is not listed in the formulary, the organisation shall have a process of evaluation, authorisation and ratification and decision-making on its subsequent inclusion in the formulary if necessary.

Standard

MOM.3.

Medications are stored appropriately and are available where required.

Objective Elements

CORE

- a. Medications are stored in a clean, safe and secure environment while incorporating the manufacturer's recommendation(s).**

Interpretation: The medication storage space shall be clean, safe and secure. The organisation shall adhere to the storage requirements of drugs as specified by the manufacturer. In the absence of manufacturer's instructions, the organisation shall develop and implement storage requirements. Storage requirements shall apply to all areas where medications are stored, including clinical areas.

Medications shall be protected from loss or theft throughout the organization. Some of the ways of ensuring this is to limit access to medication storage areas to authorised team members, locking medication carts and never leaving them unattended, or storing medications in an area that is continuously staffed. It is preferable that the medication storage area is organised. Overall cleanliness of the storage area shall be maintained.

Vaccines shall be stored at the required temperature as per manufacturers recommendations. Where appropriate, temperature monitoring of the room, the cold storage area/refrigerator shall be done at least once a day. In case of areas which are not open all days, it shall be done on all working days.

Beyond expiry date drugs (before disposal), shall be stored separately and away from drugs which are intended for patient use.

Commitment b. Sound inventory control practices guide the storage of medications throughout the organisation.

Interpretation: The Organisation shall follow inventory control practices like ABC, VED, FSN, First Expiry First Out, lead time analysis, etc. or a combination of these. The medicines could be stored in an alphabetical order of generic name.

To check for loss or theft, the organisation could conduct stock verification audits at regular intervals (as defined by the organisation) to verify the inventory and detect instances of loss or theft.

The organisation also has a mechanism for handling medications which are not part of the regular inventory. For example, not for sale medications including physician's samples.

CORE c. The organisation defines and updates its list of high-risk medication(s). *

Interpretation: The organisation shall define and update its list of high risk medications periodically. High risk/high alert medications carry a heightened risk for adverse outcomes and catastrophic harm whenever there is an error. High-risk medications/high alert medications include medicines with low therapeutic window, controlled substances, psychotherapeutic medications, look-alike and sound-alike medications, and concentrated electrolytes. The list shall be available in the pharmacy and all clinical areas where high-risk medications are stored.

Achievement d. High-risk medications are stored in areas of the organisation where it is clinically necessary.

Interpretation: High-risk medications are stored in pre-determined areas of the organisation for example, certain wards, OT, ICU. Clinical needs shall determine the availability of relevant high-risk drugs in such areas. Where applicable, (narcotics), it shall be guided by regulations. In all such areas, safeguards shall be in place to prevent inadvertent administration.

CORE

- e. **High-risk medications including look-alike, sound-alike medications and different concentrations of the same medication are stored physically apart from each other. ***

Interpretation: Many drugs in ampoules, vials or tablets may look-alike or sound-alike. These are identified periodically, and the Look-alike Sound-alike medications (LASA) list shall be made available in all units where drugs are stored. Different concentrations of the same drug need to be identified. The list shall be developed from the hospital formulary. The list will have to be revised at regular intervals depending on the changes in the formulary and changes in the packaging (in case of look-alike).

A good practice is to store the two identified look-alike/sound-alike drugs and/or different concentrations of the same drugs as far apart physically as is possible. . This is in addition to regular storage practices. In addition to the pharmacy, these storage practices shall be followed in patient care areas.

- Commitment f. The list of emergency medications is defined and is stored uniformly. ***

Interpretation: The list of emergency medications shall be prepared in consonance with sound clinical practices and documented. The list of drugs could be modified according to the needs of the clinical department for example ICU, physiotherapy, emergency, CATH lab etc. A crash cart would help the organisation to store these medications, i.e. the rows and drawers have defined medicines.

No other drug shall be kept stored with emergency medications.

CORE

- g. **Emergency medications are available all the time and are replenished promptly when used.**

Interpretation: Adequate quantity of emergency medicines shall be stocked at all times. An inventory check shall be done at least daily to ensure this. In case the organisation follows a system of sealing the emergency cart, then the check shall be carried out after each use of the cart/once every month.

Refer to COP5.c.

Standard

MOM.4. Medications are prescribed safely and rationally.

Objective Elements

- Commitment a. Medication prescription is in consonance with good practices / guidelines for rational prescription of medications. ***

Interpretation: This shall address both outpatient and in-patient prescription.

The organisation shall ensure that clinicians are trained/sensitised on the rational prescription of medications.

WHO states: "Rational use of medicines requires that patients receive medications appropriate to their clinical needs, in doses that meet their own individual requirements, for an adequate period of time, and at the lowest cost to them and their community."

Code of Medical Ethics-2002" published by Medical Council of India erstwhile National Medical Commission shall be followed.

Refer to the glossary for a definition of "prescription".

CORE

- b. The organisation adheres to the determined minimum requirements of a prescription. ***

Interpretation: Prescriptions generated within the organisation (IPD, OPD and emergency) shall be in accordance with national/international guidelines and regulatory bodies. At a minimum, the prescription shall have the name of the patient; unique hospital number; name of the drug (generic composition is mandatory except in the case of combinations of vitamins and/or minerals), strength, dosage instruction, duration and total quantity of the medicine; name, signature and registration number of the prescribing doctor. Error-prone abbreviations shall not be used. All prescriptions shall be written in capital letters. It is preferable to use digital prescription system to reduce errors.

Prescription errors or illegible prescriptions will be initialled after a single strikethrough and rewritten.

Good references are the Drugs and Cosmetics Act and the Code of Medical Ethics and Institution for Safe Medication Practices guidelines.

Commitment c. Drug allergies and previous adverse drug reactions are ascertained before prescribing.

Interpretation: Drug allergy and previous adverse drug reaction shall be ascertained during the initial consultation, before prescribing or at any point of time during care. It is a good practice to document drug allergies prominently in the medical record, both in OP and IP.

Excellence d. The organisation has a mechanism to assist the clinician in prescribing appropriate medication.

Interpretation: The organisation needs to provide its clinicians with a mechanism(s) to help identify drug interactions, food-drug interactions, alcohol-drug interactions, therapeutic duplication, dose adjustments etc. This could either be in electronic or in the physical form.

CORE e. Reconciliation of medications occurs at transition points of patient care.

Interpretation: The purpose of medication reconciliation is to ensure that the list of medications that a patient has to receive is complete and up to date with past clinical conditions and present care plan.

The prescribed medications shall be checked for accuracy at the transition points, such as the time of admission, either direct or after admission from the emergency, transfer of the patient from one ward setting/department to another, or at the time of discharge. It is preferable that medication reconciliation also occurs after cross-consultation. Medication reconciliation shall be documented. There is a system for effective communication during handover regarding the reconciliation of medications.

CORE f. Verbal orders are implemented by ensuring safe medication management practices. *

Interpretation: The organisation shall ensure safe medication management practices through written guidance and implementation of the same. The written guidance shall mention who can give a verbal order, when can it be given and how the order will be authenticated. Verbal orders shall be limited to urgent situations where immediate written or electronic communication is not practical. To the extent possible, their usage shall be limited. The organisation shall have an approved list of formulary drugs which can be ordered verbally. This list can be defined either by inclusion or exclusion.

It shall ensure that the procedure incorporates good practices like “repeat back/read back”.

A verbal order shall be counter-signed by the doctor who ordered it within 24 hours of ordering.

For the definition of ‘verbal order’, refer to the glossary.

Achievement g. Audit of medication orders / prescription is carried out to check for safe and rational prescription of medications.

Interpretation: The scope of the audit shall include:

- legibility, use of capitals in written orders;
- the appropriateness of the drug, dose, frequency, and route of administration;
- the presence of therapeutic duplication;
- the possibility of drug interaction and measures taken to avoid the same;
- the possibility of food-drug interaction and measures taken to avoid the same.
- scope for dosage adjustment in renal and hepatic impairment.
- the possibility of intravenous incompatibility.
- presence of inappropriate dilutions and duration of medications Infusions.
- the requirements of this standard (MOM 4b, e-h).

This shall be done at least once a month using a representative sample size.

It could preferably be done by a clinical pharmacologist/clinical pharmacist. In case there is no clinical pharmacologist/clinical pharmacist, it could be done by a multidisciplinary committee/team, who is trained in audit of medication orders/prescriptions. It is preferable that this is done for all prescriptions as ‘live audit’ before the medicines are dispensed.

Achievement h. Corrective and / or preventive action(s) is taken based on the audit, where appropriate.

Interpretation: Where appropriate, a corrective and/or preventive action(s) is taken based on the root-cause analysis. The records of the same shall be maintained.

Standard

MOM.5.

Medications orders are written in a uniform manner.

Objective Elements

Commitment a. The organisation ensures that only authorised personnel write orders. *

Interpretation: Medication orders shall be written by a doctor who at a minimum, holds an MBBS qualification. In case there is any other category of staff authorised to write medication orders, the same shall be backed by a legislation or government order. The medication order card in the IP shall have the orders written by a doctor, even if it is the case of transcribing orders of the treating consultant from an OP record or an admission note. In facilities which use Electronic Medical Record (EMR), the doctor shall directly enter the prescription in the Hospital Information System (HIS) using his or her unique login. In case the HIS entry is made by an assistant, the same shall be verified and authorised by the doctor.

Commitment b. Medication orders are written in a uniform location in the medical records, which also include the patient's name and unique identification number.

Interpretation: All the orders for medicines shall be recorded on a uniform location of the medical record. Only medications written in this location shall be administered to the patient. It is imperative that medication orders that are written in any other location of the medical record be moved to this location, electronic orders, when typed, shall again follow the same principles. It is preferable that the prescription and the administration record is on the same sheet. This would help minimise medication errors. A drug 'Kardex' could be used for this purpose. The treatment orders are written daily or authorised daily in a 'Kardex' like format. Phrases like "CST"/"continue same treatment"/"repeat all"/"repeat 1,4,5,8" shall not be accepted.

Whenever there is a modification in the medication order in the existing order for a particular drug, a fresh order will have to be written for that drug for example, Tab. Paracetamol 500 mg QID changed to Tab. Paracetamol 500 mg BD – this shall warrant the first order to be discontinued and a fresh medication order to be written. A strike-through or over-writing of the previous order is not acceptable.

Commitment c. Medication orders are legible, dated, timed and signed.

Interpretation: All hand-written medication orders shall be written in capital letters. In case abbreviations are used, a list of approved standardised abbreviations for medication orders shall be used throughout the organisation. Error-prone abbreviations shall not be used. A good reference is the Institution for Safe Medication Practices guidelines.

The identity of the person who has written the medication order shall be traceable. This could be done by either writing the name against every order or by having a 'master signature list' in the medical record which has the name of the person against the signature or by stating the employee code number against every medication order.

Commitment d. Medication orders contain the name of the medicine, route of administration, strength to be administered and frequency / time of administration.

Interpretation: In case of medication orders which have two or more drugs (tablet/capsule/injection) the strength of all the individual drugs shall be written. This may not be applicable for preparations having a combination of vitamins and/or minerals. Medication orders shall be recorded separately if the strength differs for each time of administration.

There shall be a mechanism for taking action when medications orders are incomplete with respect to any of the above parameters.

Refer to MOM.7.d

Standard

MOM.6.

Medications are dispensed in a safe manner.

Objective Elements

Commitment a. Dispensing of medications is done safely. *

Interpretation: Written guidance shall be laid down for the safe dispensing of medication. Medications shall be dispensed only against a valid prescription or medication order (except for over-the-counter drugs). Medication shall be

checked before dispensing and include a check of the generic composition, formulation, expiry date, and where applicable, the strength.

This shall include both bulk and retail pharmacy.

Physicians' samples shall not be sold.

Commitment b. Medication recalls are handled effectively. *

Interpretation: The organization shall have an established system for medication recalls. Recall may be based on communication from regulatory authorities, manufacturer or internal feedback (for example, visible contaminant in IV fluid bottle). The recall procedure in response to internal feedback also includes providing information to the appropriate regulatory authority. A record of the same shall be maintained on occurrence.

Refer to MOM.1.e.

Commitment c. Near-expiry medications are handled effectively. *

Interpretation: The organisation could define as to what constitutes "near expiry" medications, for example, three months before the expiry date. The organisation's mechanism shall ensure that near expiry medications are withdrawn and that no beyond expiry date medication is available.

CORE d. Dispensed medications are labelled. *

Interpretation: At a minimum, the label shall include the dosage instruction in a manner that the patient understands. Labelling is applicable only for out-patients. In instances when medicines are dispensed either as cut strips or from bulk containers, the label must include the drug name, strength, dosage instruction (in a manner that the patient understands) and expiry date. The organisation could consider using technology (for example, QR codes on individual medicines) to optimise the process. This shall be applicable for both in-patients and out-patients. It shall also be applicable where drugs are reconstituted for example Chemotherapy drugs.

CORE

e. High-risk medication orders are verified before dispensing.

Interpretation: High-risk medications shall be given only after written orders, and it shall be verified by the staff before dispensing. This shall adhere to statutory requirements, where applicable.

Commitment

f. Return of medications to the pharmacy is addressed. *

Interpretation: The organisation shall have written guidance to direct the return of medications to the pharmacy. At a minimum, the written guidance shall address the list of medications which would be accepted for return (either by inclusion or exclusion) and minimum conditions to be met for return of medications. It is preferable that any drug which has a specific temperature storage requirement not be accepted for return once it has gone out of the organisation's premises. The minimum conditions to be met for return of medications could include the drug name, strength, batch number and expiry date matching the bill; no visible damage etc. It is preferable that the organisation creates awareness regarding the return of medications. Some ways of doing this are having a signage stating the salient points mentioned above, putting up this information on the website, printing the information on the receipt etc.

Standard

MOM.7.

Medications are administered safely.

Objective Elements

Commitment

a. Medications are administered by those who are permitted by law to do so.

Interpretation: Only a registered nurse or doctor with a minimum of MBBS qualification shall administer medication. In case there is any other category of staff authorised to administer medication, a legislation or government order shall back the same.

Commitment b. Prepared medication is labelled before preparation of a second drug.

Interpretation: Labelling is required when more than one drug is prepared and loaded. Examples of these are anaesthetic drug preparation in OTs, chemotherapy drugs etc.

Commitment c. The patient is identified before administration.

Interpretation: At a minimum, two identifiers shall be used for patient identification, with one of them being the unique identification number (for example hospital number/IP number, etc.) and full name of the patient.

Refer to COP.1.b.

CORE

d. Medication is verified from the medication order and physically inspected before administration.

Interpretation: Staff administering medications shall verify the medication order and ensure that medications are administered appropriately. The general appearance of the medication (for example melting, clumping, etc.) and the expiry dates shall be checked before administration. If any of the parameters concerning an order, such as name, strength, route or frequency or time are missing/incomplete, the medication administration shall be deferred, pending early verification by the treating team. In case the confirmation is obtained verbally, it shall be considered a verbal order and the procedure for verbal orders shall be adhered to.

In case of high-risk medication(s), the verification shall be done by at least two staff (nurse-nurse or nurse-doctor), independently and documented.

The nurses shall be knowledgeable regarding high-risk medications and shall be empowered to highlight prescription errors noted while verifying the orders.

Commitment e. Strength is verified from the order before administration.

Interpretation: The person administering the drug shall verify the strength from the medication order before administration. In case of discrepancy, medication administration shall be deferred.

Refer to MOM.7.d

Commitment f. The route is verified from the order before administration.

Interpretation: Where applicable, the site/ route of administration shall also be verified.

Refer to MOM.7.d

Commitment g. Timing is verified from the order before administration.

Interpretation: The organisation shall have documentation to support the time of administration of drugs for which the time has not been written. For example, 1-1-1, BD etc. The suggested timings for these medicines have to be adhered. "ISMP Acute Care Guidelines for Timely Administration of Scheduled Medications" provide guidance on scheduling medications and classifies them into time-critical and non-time-critical. The organisation could adopt/adapt the same.

CORE**h. Measures to avoid catheter and tubing mis-connections during medication administration are implemented. ***

Interpretation: The organisation ensures that inadvertent administration of a drug through the wrong route is avoided. This could be done by using design features that prevent misconnections and prompt the user to take the correct action. Intravenous (IV) extension tubes shall not be used for epidurals, irrigation, drains, and central lines or to extend enteric feeding tubes. Care shall be taken to position functionally dissimilar tubes used in patient care away from one another. Staff administering medications shall trace all lines from their origin to the connection port to verify attachments before making any connections or reconnections, or administering medications, solutions, or other products.

Commitment i. Medication administration is documented.

Interpretation: The organisation shall ensure that documentation of medication administration is done in a uniform location. It shall include the name of the medication, strength, route of administration, timing and the name/employee ID number and signature of the person who has administered the medication. Medicines administered shall be documented each time for each dose of the same medication separately. The records shall reflect the actual administration. For example, if brand Y was given in place of brand X (same generically), the documentation shall be of brand Y. Similarly, if the order was for a tablet of 250 mg, but the administration was ½ a tablet of 500 mg, the latter shall be documented. In the case of infusions, it shall capture the start time, the rate/volume of infusion and end time. In case of continuous infusion, the drop rate/volume shall be documented and the total volume infused shall be calculated for each shift.

Commitment j. Measures to govern patient's self-administration of medications are implemented. *

Interpretation: At the outset, the organisation could define if it would permit the self-administration of medications. In case the organisation permits the same, the written guidance shall define the medications which the patient can self-administer. It is preferable that the organisation also incorporates a method for assisting self-administration of medications. The organization shall ensure that the patient is reminded to take the medication (before every dose) and document the same. For example, self-administration of insulin.

Commitment k. Measures to govern patient's medications brought from outside the organisation shall be implemented. *

Interpretation: At the outset, the organisation could define if it would permit the patient getting his/her medications. In case of the organisation permitting the same, written guidance shall include the pre-requisites for such a medication (for example, clear label with mention of the name, strength, expiry date, batch number etc.).

Standard

MOM.8.

Patients are monitored after medication administration.

Objective Elements

Commitment a. Patients shall be monitored after medication administration. *

Interpretation: Relevant monitoring shall be done collaboratively to verify that the medicine is having its intended effect. It could also include monitoring the effects of medications (beneficial or adverse) through laboratory results. Besides, this shall help identify near misses, medication errors and adverse drug reactions.

The organisation shall define those situations and medications where more frequent monitoring is required. For example, administration of high-risk medicines. The effect of medication in high risk patients like those on dialysis, in the ICU and elderly group shall be monitored on a regular basis.

Commitment b. Medications shall be changed based on the monitoring where appropriate.

Interpretation: Medication changes are based on clinical response and adverse drug reactions, if any.

CORE

c. The organisation shall capture near misses, medication errors and adverse drug reactions. *

Interpretation: Near misses, medication errors and adverse drug reactions shall be defined. This shall be in consonance with best practices. The organization shall have a written guidance on the process to capture near misses, medication errors and adverse drug reactions. This shall incorporate identifying, documenting, reporting, analysing and action taken regarding near misses, medication errors and adverse drug reactions.

Refer to the glossary for “near miss”, “medication error” and “adverse drug reaction”.

Commitment d. Near misses, medication errors and adverse drug reactions shall be reported within a specified time frame. *

Interpretation: The organisation shall define the timeframe for reporting once any of this has occurred and adhere to the same.

Commitment e. Near misses, medication errors and adverse drug reactions are collected and analysed.

Interpretation: Details of the near misses, medication errors and adverse drug reactions are collected and analysed by the multidisciplinary committee. The analysis shall be completed in a defined time frame. It is preferable that a clinical pharmacologist/clinical pharmacist be a part of this exercise.

Commitment f. Corrective and / or preventive action(s) are taken based on the analysis.

Interpretation: Where appropriate, corrective and/or preventive action are taken. The records of the same shall be maintained. It is preferable that corrective and/or preventive action(s) be taken based on the root-cause analysis.

Standard

MOM.9.

Narcotic drugs and psychotropic substances, chemotherapeutic agents and radio-pharmaceuticals are used in a safe manner.

Objective Elements

- Commitment a. Narcotic drugs and psychotropic substances, chemotherapeutic agents and radio-pharmaceuticals are used safely. ***

Interpretation: Written guidance, developed in consonance with local and national regulations/guidelines shall be implemented. The written guidance could address all the objective elements of this standard. Examples of regulations/guidelines are Narcotic Drugs and Psychotropic Substances Act and AERB guidelines.

- Commitment b. Narcotic drugs and psychotropic substances, chemotherapeutic agents and radio-pharmaceuticals are prescribed by appropriate caregivers.**

Interpretation: For narcotic drugs, it shall be the designated/ privileged medical officer(s).

Chemotherapy shall be prescribed by those who have the knowledge to monitor and treat the adverse effect of chemotherapy. This is preferably a medical oncologist or a doctor who has been trained and has achieved competency in the same.

Radiopharmaceuticals shall be prescribed only by a caregiver authorised by the statutory body.

- Commitment c. Narcotic drugs and psychotropic substances, chemotherapeutic agents and radio-pharmaceuticals drugs shall be stored securely.**

Interpretation: Narcotic drugs shall be stored securely in consonance with statutory requirements. The security measures shall ensure that these medications are not diverted and abused.

Chemotherapeutic agents shall be accessible only to authorised personnel.

Radiopharmaceuticals shall be labeled and stored as per AERB guidelines.

It is preferable that these medications are stored separately from other medications.

Commitment d. Chemotherapy and radio-pharmaceuticals shall be prepared properly and safely and administered by qualified personnel.

Interpretation: It is required that qualified personnel have received special training in the preparation and administration of chemotherapeutic drugs.

A bio-safety cabinet of class II (preferably IIA) with appropriate personal protective equipment shall be used for preparing/mixing chemotherapeutic drugs.

Radio-pharmaceuticals shall be prepared and administered by an authorised caregiver.

Commitment e. A proper record shall be kept of the usage, administration and disposal of narcotic drugs and psychotropic substances, chemotherapeutic agents and radio-pharmaceuticals.

Interpretation: A strict inventory control shall be kept for narcotic drugs and psychotropic substances, chemotherapeutic agents and radiopharmaceuticals. Record of usage, administration, wastage and disposal of narcotic drugs shall be kept following statutory requirements. These shall be disposed off according to existing statutory requirements (including Narcotic Drugs and Psychotropic Substances Act, AERB rules and Biomedical waste management rules) and the manufacturer's recommendation (where applicable).

Standard

MOM.10.

Implantable prosthesis and medical devices shall be used in accordance with laid down criteria.

Objective Elements

Commitment a. Usage of implantable prosthesis and medical devices is guided by scientific criteria for each item and national / international recognised guidelines / approvals for such specific item(s).

Interpretation: The organisation shall ensure that relevant and sufficient scientific data are available before selection. It shall also look for any international (for example US-FDA) or national notification or applicable regulations (Central Drugs Standard Control Organization notification based on the Drugs and Cosmetics Act) for approval of the particular product. The multidisciplinary committee shall be responsible for approving the use of a particular implant.

Commitment b. The organisation implements a mechanism for the usage of the implantable prosthesis and medical devices. *

Interpretation: The organisation shall have written guidance to direct procurement, storage/stocking, issuing and usage of implantable prosthesis and medical devices. Infection prevention and control requirement related to use of prosthesis and medical devices is implemented. This shall address statutory regulations/guidelines and manufacturer's recommendation(s).

Commitment c. Patient and his / her family are counselled for the usage of the implantable prosthesis and medical device, including precautions if any.

Interpretation: Precautions could include non-usage of specific drugs and reporting to the hospital if a particular symptom occurs. The organization shall document the details of counselling provided to the patient/ his or her family in informed consent.

Commitment d. The batch and the serial number of the implantable prosthesis and medical devices are recorded in the patient's medical record, the master logbook and the discharge summary.

Interpretation: In case where implantable prosthesis do not have pre-labelled stickers, the organisation shall have suitable mechanisms in place for identifying the implant (manufacturer, type, size, batch number, serial number and any other important detail).

Achievement e. Process of recall of implantable prosthesis and medical devices are handled effectively. *

Interpretation: Process of recall may be based on communication from regulatory authorities, manufacturer or internal feedback. Recall procedure in response to internal feedback also includes providing information to appropriate regulatory authority and the manufacturer. The record shall be maintained on occurrence.

Refer to MOM.1.e. and FMS.5.g.

Standard

MOM.11.

Medical supplies and consumables are stored appropriately and are available where required.

Objective Elements

- Commitment a. The organisation adheres to the defined process for the acquisition of medical supplies and consumables. ***

Interpretation: In this context, medication supplies and consumables refer to items which are used in patient care excluding medications and implants. The process shall address the issues of vendor selection, vendor evaluation, indenting process, generation of the purchase order and receipt of goods.

- Commitment b. Medical supplies and consumables are used in a safe manner, where appropriate.**

Interpretation: The items shall be opened and used using relevant precautions to maintain sterility and integrity.

- Commitment c. Medical supplies and consumables are stored in a clean, safe and secure environment; incorporating the manufacturer's recommendation(s).**

Interpretation: The organisation shall ensure that the storage requirements specified by the manufacturer are adhered to. This shall apply to all areas where these are stored, including wards. They shall be protected from loss or theft. Overall cleanliness of the storage area shall be maintained. Hazardous materials shall be identified and kept safely.

- Commitment d. Sound inventory control practices guide storage of medical supplies and consumables.**

Interpretation: The organisation shall follow or demonstrate ABC, VED, FSN, First Expiry First Out, lead time analysis, etc.

Commitment e. There is a mechanism in place to verify the condition of medical supplies and consumables.

Interpretation: Medical supplies and consumables shall be in a condition suitable for safe usage. The condition of these materials shall be checked before dispensing and usage. For example, opened up package, damp cotton roll, physical damage, and discolouration.

References

1. US Department of Health and Human Services. Agency for Healthcare Research and Quality. Patient Safety Network. Patient safety primer: never events. 2011. [cited 2023 Dec 5] Available on: <https://psnet.ahrq.gov/issue/serious-reportable-events-healthcare-2011-update>.
2. Kalaiselvan V, Srivastava S, Singh A, Gupta SK. Pharmacovigilance in India: present scenario and future challenges. *Drug safety*. 2019 Mar 7;42:339-46.
3. Whaley C. Perspectives from pharmacists, prescribers and patients on sharing medication reason for use (Master's thesis, University of Waterloo). [cited 2023 Dec 5] Available from : <https://uwspace.uwaterloo.ca/handle/10012/15817>.
4. MacMaster HW, Gonzalez S, Maruoka A, San Luis C, Stannard D, Rushakoff JA, Rushakoff RJ. Development and implementation of a subcutaneous insulin pen label bar code scanning protocol to prevent wrong-patient insulin pen errors. *The Joint Commission Journal on Quality and Patient Safety*. 2019 May 1;45(5):380-6.
5. Institute for Safe Medication Practices. FDA and ISMP lists of look-alike drug names with recommended tall man letters. [cited 2023 Dec 5] Available from: <https://www.ismp.org/sites/default/files/attachments/2017-11/tallmanletters.pdf>.
6. Jungreithmayr V, Meid AD, Haefeli WE, Seidling HM. The impact of a computerized physician order entry system implementation on 20 different criteria of medication documentation—a before-and-after study. *BMC Medical Informatics and Decision Making*. 2021 Dec;21:1-2.
7. Schepel L, Lehtonen L, Airaksinen M, Lapatto-Reiniluoto O. How to identify organizational high-alert medications. *Journal of patient safety*. 2021 Dec 1;17(8):e1358-63.
8. Jayasinghe M, Srilal TL, Subasinghe S, Zawahir S. Identification of Confusing Medicine Proprietary Names: Toward Safe Medicine Use—A Cross-Sectional Study in Sri Lanka. *Therapeutic Innovation & Regulatory Science*. 2023 Nov;57(6):1248-59.
9. Kahn S, Abramson EL. What is new in paediatric medication safety. *Archives of Disease in Childhood*. 2019 Jun 1;104(6):596-9.
10. National Coordinating Council for Medication Error Reporting and Prevention. (2015, January 29). Recommendations to Enhance Accuracy of Prescription/Medication Order Writing. [cited 2023 Dec 5] Available from: <https://www.nccmerp.org/recommendations-enhance-accuracy-prescription-writing>
11. National Coordinating Council for Medication Error Reporting and Prevention. (2015, September 2). Recommendations to Reduce Medication Errors Associated with Verbal Medication Orders and Prescriptions. [cited 2023 Dec 5]. Available from: <https://www.nccmerp.org/recommendations-reduce-medication-errors-associated-verbal-medication-orders-and-prescriptions>.
12. National Coordinating Council for Medication Error Reporting and Prevention. (2015, September 1). Recommendations to Enhance Accuracy of Dispensing Medications. Recommendations to Enhance Accuracy of Dispensing Medications. [cited 2023 Dec 5]. Available from : <https://www.nccmerp.org/recommendations-enhance-accuracy-dispensing-medications>.
13. National Coordinating Council for Medication Error Reporting and Prevention. (2015, January 29). Recommendations to Enhance Accuracy of Prescription/Medication Order Writing. Available from: <https://www.nccmerp.org/recommendations-enhance-accuracy-prescription-writing> [cited 2023 Dec 5].

14. National Coordinating Council for Medication Error Reporting and Prevention. (2015, September 2). Recommendations to Enhance Accuracy of Administration of Medications. https://www.nccmerp.org/search/site?search_api_fulltext=Recommendations+to+Enhance+Accuracy+of+Prescription%2FMedication+Order+Writing.+Retrieved. [cited 2023 Dec 5].
15. National Coordinating Council for Medication Error Reporting and Prevention. (2015, January 30). About Medication Errors. About Medication Errors. [cited 2023 Dec 5]. Available from: https://www.nccmerp.org/search/site?search_api_fulltext=About+Medication+Errors.
16. Tully AP, Hammond DA, Li C, Jarrell AS, Kruer RM. Evaluation of medication errors at the transition of care from an ICU to non-ICU location. *Critical care medicine*. 2019 Apr 1;47(4):543-9.
17. Nyabuti AO. Core Indicators Of Appropriate Drug Use At Public Primary Healthcare Centers In Kisii County, Kenya (Doctoral dissertation, University of Nairobi). [cited 2023 Dec 5]. Available from : <http://erepository.uonbi.ac.ke/handle/11295/152925>.
18. World Health Organization. (n.d.). Improving Medication safety. Available from : <https://www.who.int/patientsafety/solutions/patientsafety/PS-Solution7.pdf?ua=1>. [cited 2023 Dec 5].
19. World Health Organization. (n.d.). Look-Alike, Sound-Alike Medication Names. [cited 2023 Dec 5]. Available from: <https://www.who.int/patientsafety/solutions/patientsafety/PS-Solution1.pdf?ua=1>
20. World Health Organization. (n.d.). Rational use of medicines. Available from: <https://www.who.int/medicines/areas/rational>[cited 2023 Dec 5].
21. World Health Organization. (n.d.). The High 5s Project –Standard Operating Protocol Assuring Medication Accuracy at Transitions in Care: Medication Reconciliation. [cited 2023 Dec 5]. Available from: <https://www.who.int/patientsafety/implementation/solutions/high5s/h5s-sop.pdf>.
22. World Health Organization. (n.d.). WHO Model Lists of Essential Medicines. [cited 2023 Dec 5] Available from: <https://www.who.int/medicines/publications/essentialmedicines/en/>.
23. Bryan R, Aronson JK, Williams A, Jordan S. The problem of look-alike, sound-alike name errors: Drivers and solutions. *British Journal of Clinical Pharmacology*. 2021 Feb;87(2):386-94.
24. Barnsteiner JH. Medication Reconciliation. Agency for Healthcare Research and Quality, AHRQ, 08, 1–1403.
25. Wilson L. Administration of Medications in Wisconsin Schools. Wisconsin Department of Public Instruction. 2022 Apr.
26. Mirza M, Verma M, Sahoo SS, Roy S, Kakkar R, Singh DK. India's Multi-Sectoral Response to Oxygen Surge Demand during COVID-19 Pandemic: A Scoping Review. *Indian Journal of Community Medicine: Official Publication of Indian Association of Preventive & Social Medicine*. 2023 Jan;48(1):31.
27. Singh P, Vaishnav Y, Verma S. Development of pharmacovigilance system in India and paradigm of pharmacovigilance research: an overview. *Current Drug Safety*. 2023 Nov 1;18(4):448-64.
28. World Health Organization. The world health report 2002: reducing risks, promoting healthy life. World Health Organization; 2002. [cited 2023 Dec 5]. Available from:
29. Seden K, Kirkham JJ, Kennedy T, Lloyd M, James S, Mcmanus A, Ritchings A, Simpson J, Thornton D, Gill A, Coleman C. Cross-sectional study of prescribing errors in patients admitted to nine hospitals across North West England. *BMJ open*. 2013 Jan 1;3(1):e002036.
30. Stark HE, Graudins LV, McGuire TM, Lee CY, Duguid MJ. Implementing a sustainable medication reconciliation process in Australian hospitals: The World Health Organization High 5s project. *Research in Social and Administrative Pharmacy*. 2020 Mar 1;16(3):290-8.

31. Tully AP, Hammond DA, Li C, Jarrell AS, Kruer RM. Evaluation of medication errors at the transition of care from an ICU to non-ICU location. *Critical care medicine*. 2019 Apr 1;47(4):543-9.
32. Bourges JL, Bloquel C, Thomas A, Froussart F, Bochot A, Azan F, Gurny R, BenEzra D, Behar-Cohen F. Intraocular implants for extended drug delivery: therapeutic applications. *Advanced drug delivery reviews*. 2006 Nov 15;58(11):1182-202.
33. Lee SS, Hughes P, Ross AD, Robinson MR. Biodegradable implants for sustained drug release in the eye. *Pharmaceutical research*. 2010 Oct;27:2043-53.
34. Zilberman M, Kraitzer A, Grinberg O, Elsner JJ. Drug-eluting medical implants. *Drug Delivery*. 2010:299-341.
35. Khanna M, Garg S. Narcotic drugs and psychotropic substances act (ndps 1985): a critical analysis. *Nimitmai Review Journal*. 2020 Jun 24;3(1):28-32.
36. Singhal K, Jain R, Bhardwaj C. Enforcement of the Narcotics Drugs and Psychotropic Substances Act: Reality Far from Ideal?. *Indian JL & Legal Rsch..* 2021;2:1.
37. Kalura R. Narcotic Drugs and Psychotropic Substances (NDPS) Act, 1985. Issue 4 *Int'l JL Mgmt. & Human..* 2021;4:2177.
38. Gazette Notification: Ministry of Revenue (Department of Revenue) New Delhi 25th March 2015. Narcotic Drugs and Psychotropic Substances(Second Amendment Rules), 2015. [cited 2023 Dec 5]. Available from: <https://www.cbn.nic.in/pdf/GSR426E01-07-2014.pdf>.
39. Gazette Notification: Ministry of Revenue (Department of Revenue) New Delhi 1st July 2014. The narcotic drugs and psychotropic substances (amendment) act, 2014. [cited 2023 Dec 5]. Available from: <https://dor.gov.in/sites/default/files/NDPS-Amendment%20Act%20-%202014.pdf>.
40. Gazette Notification: Ministry of Finance (Department of Revenue) New Delhi 23rd December 2022. Medical devices rules, 2017. [cited 2023 Dec 5]. Available from: https://cdsco.gov.in/opencms/resources/UploadCDSCOWeb/2022/m_device/Medical%20Devices%20Rules,%202017.pdf.
41. Government of India, Ministry of Chemicals and Fertilizers, Department of Pharmaceuticals(Public Notice dated: 01.02.2022) List of available standards of Bureau of Indian Standards for Medical Devices. [cited 2023 Dec 5]. Available from: https://pharmaceuticals.gov.in/sites/default/files/Public%20Notice_BIS%20Standards.pdf